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Impact of high-frequency harmonics (0–9 kHz) generated by grid-connected inverters on distribution transformers☆

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ABSTRACT

The level of current harmonics circulating in a transformer winding can affect its operating temperature and lifetime. Although the existing standards mainly consider the impact of harmonics up to 2 kHz, higher frequency harmonics generated by high power converters utilized in renewable energy sources can also contribute to the temperature rise of a transformer. Pulse Width Modulated (PWM) voltage generated by power converters can generate significant high-frequency harmonics at its switching frequency. The switching frequency of converters in high-power applications is mainly between 2 and 9 kHz. Consequently, in this article, the importance of also using higher frequency harmonics in calculating transformer core loss and temperature rise is investigated as per IEEE standard conclusions. For that purpose, transformer K-factor values have been calculated in simulated and practical distribution systems, in both long-term and transient conditions. The results confirm the importance of considering higher-order current harmonics (more than 2 kHz) when calculating K-factor and estimating transformer temperature rise and lifetime.

1. Introduction

Power distribution networks have faced significant changes over the last decades as more power electronic-based technologies have been employed. For instance, the transition from traditional fossil fuels to renewable energy has increased the utilization of solar panels and wind turbines, which employ power electronic inverters to connect to the grid. Moreover, power electronic devices have been utilized to improve the efficiency of loads like variable speed drives. These devices are nonlinear loads and inject harmonics into the system, which can lead to higher transformer temperatures and lower insulation residual life [1]. Utility-owned distribution and power transformers are designed to operate when the total harmonic distortion of the supply voltage is less than 5%, and the load current harmonics content is also less than 5% of its rated value, as mentioned in IEC 60076-1 [2]. This standard mentions that excessive harmonic loading can cause temperature rise higher than the rated operating temperature. This can lead to lower residual life of the insulation than expected, which could be a serious concern for a utility. An author in [3] has shown that in load cases with high harmonic content like a drilling rig, a transformer could experience overloading as high as twice the normal operating condition. This observation has been based on higher ohmic, eddy-current, and eddy current losses at the presence of harmonic loads.

When harmonic loads are present, a possible strategy to use existing utility transformers is to derate them, and then plan replacement once the load nears this derating. This method, however, requires information on existing harmonics and the transformer model. Another method to extend the lifetime of distribution under the harmonic load condition is to control its tap setting based on volt-var optimization [4]. Authors in [4] have evaluated this method considering the impact of harmonic contents of different load types; however, only harmonics contents up to 19th have been considered in the calculations.

To improve the efficiency of inverters, manufacturers are designing them to operate at faster switching frequencies. Although it has been usual for only harmonics up to 2 kHz to be investigated, these higher switching frequencies have created a need to investigate the effects of

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harmonics in higher frequencies. The effect of harmonics on transformer losses and lifetime has concerned some researchers to examine the required derating under nonlinear loads [5–11]. In [5], load analysis of a building to calculate the losses caused by nonlinear loads has been conducted. The results show that transformer losses caused by harmonics can be more than double compared with when only linear loads are present. Similarly, the measurement of eddy current losses and comparison with harmonic losses factor (F_{HL}) for a transformer is presented in [6]. The results show that eddy current losses are proportional to the square of the frequency. Authors in [8] have used a modified nonlinear transformer model to calculate the transformer derating. Unlike the IEEE C57.110 standard [12], where the derating relies only on winding losses (current harmonics), the proposed derating considers the effect of voltage harmonics and unbalance on core losses. The abovementioned articles, however, have not investigated the impact of high-frequency harmonics.

A special type of transformer called K-rated transformer is recommended to be used when loads generate a high level of current harmonics [13]. This type of transformer is equipped with larger size neutral wiring and core material with lower loss. To find the right value of the K-rated transformer, the target load current harmonics need to be first evaluated. Although different K-rated transformers are commercially available, their large-scale deployment has not yet been considered due to higher cost. Currently, utilities only use harmonics less than 50th order in calculations of required K-rated transformer. However, higher frequency harmonics also need to be included to have a more accurate estimation. Authors in [14] have mentioned that K-factor is a crucial factor in evaluating the temperature rise by harmonics. Current harmonics caused by nonlinear loads can generate a higher temperature rise in the insulation, which reduces the transformer life expectancy [15,16]. A practice to establish transformer capability when supplying nonsinusoidal currents is recommended in IEEE std C57.110 [12], which can be considered as a guideline for derating power transformers to maintain their lifetime expectancy.

The effect of load current harmonics, including renewable energy systems and EV chargers on distribution transformers is studied in [17–22]. An experiment has been done in [19] to investigate the effects of grid-connected inverters on a dry-type transformer temperature and lifetime expectancy. At rated load, the low-order current harmonics below 2 kHz from the inverter raised the insulation temperature of the transformer by 1.2 °C (from a base of 210 °C), which would decrease its life expectancy at this load by 8.3%. Authors in [20] have evaluated the impact of current harmonics generated by two types of EV chargers on power loss and temperature rise of distribution transformers. Using harmonic loss factor (F_{HL}) the presented simulation results show that the expected lifetime of the transformer could decrease significantly if the EV charger load level is higher than 80% of the rated load. Similarly, authors in [21] have evaluated the impact of distorted load currents in an industrial system on transformer temperature rise and lifetime. The simulation results show that with a nonlinear load of around 80%, the transformer aging acceleration factor (F_{AA}) can be around 2.97, showing significant reduction in a lifetime (less than half of the expected value).

Transformer losses caused by nonlinear loads can increase the temperature of insulation, which reduces its residual life. Therefore, in the presence of nonlinear loads, a transformer may need to be derated. However, the studied literature presented above has not considered the impact of high order current harmonics (above 2 kHz) on transformer losses, which could be significant due to the increasing level of power electronics circuits [23]. Moreover, they could have a higher impact on the transformer compared to low-order harmonics as they tend to pass through the edge of the conductor. Consequently, the main aim of this work is to present the possibility of a significant impact of higher frequency current harmonics on the transformer loss, which is confirmed with several practical measurements. In the next section, the importance of addressing 2–9 kHz current harmonics and under-progress standardization are discussed. To evaluate the impact of harmonics on transformer, its mathematical loss modelling is then presented in section III. Simulation, field and lab measurements of current harmonics are presented in section IV for different systems with different loading conditions. Additionally, to compare the impact of low-order harmonics (below 2 kHz) with high-order harmonics, some indicative parameters extracted from K_f are proposed in that section. In the next step, the expected temperature rise of the transformer is calculated for cases without and with current harmonics in section V. Section VI then provides the main discussions related to the practical results. Finally, section VII concludes the paper. The main contributions of this research can be summarized as follows:

- highlighting the importance of high-frequency harmonics – above 2 kHz in distribution networks.
- developing new K-factor indices to address harmonics within 2–9 kHz
- analyzing the impacts of 2–9 kHz harmonics on transformer temperature rise and lifetime compared to the conventional frequency range of 0–2 kHz

underlining the requirement to measure high-frequency harmonics in the wind and solar farms, where a high number of grid-connected inverters are connected to distribution transformers and may affect their lifetime.

2. Addressing emerging harmonics at 2–9 kHz range

As shown in Fig. 1, the existing international technical documents and standards cover all grid-connected electrical and electronic systems for the frequency ranges of (0–2 kHz) and (above 150 kHz) based on IEC 61000-3-2, IEC 61000-3-12, IEEE 519, and CISPR (International Special Committee on Radio Interference), respectively. However, there are currently no immunity and emission limits and a measurement method at product and system levels for harmonics within the frequency ranges of 2–9 kHz and 9–150 kHz, which are the most important issues in the international standardization committee (IEC, TC77A). IEC-TC77A-WG1 and WG8 have been working on the definition of compatibility levels for the frequency ranges of 2–9 kHz and 9–150 kHz. Fig. 1 shows a broad range of frequency, where existing and future international regulations for each frequency range are addressed.

Fig. 2 shows two different power converter topologies – diode rectifier and Active Front End (AFE), which are utilised in many high-power applications such as motor drives and renewable energy systems. The power converter based on the diode rectifier can have a large or small DC-link capacitor which can generate low or high-frequency resonance with the grid impedance. The AFE converter generates high-frequency harmonics at its switching frequency, and it is sensitive to grid impedance, background harmonics and disturbances which can

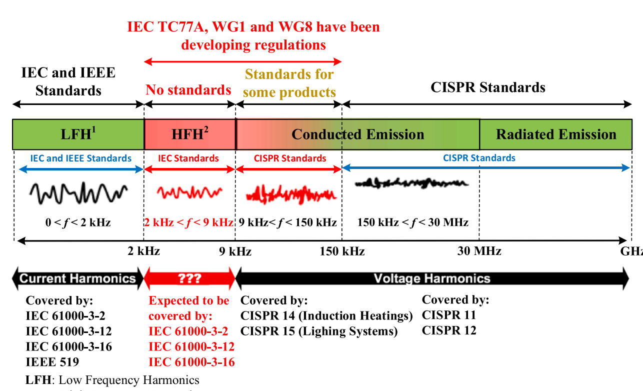


Fig. 1. International regulations for low- and high-frequency harmonics.

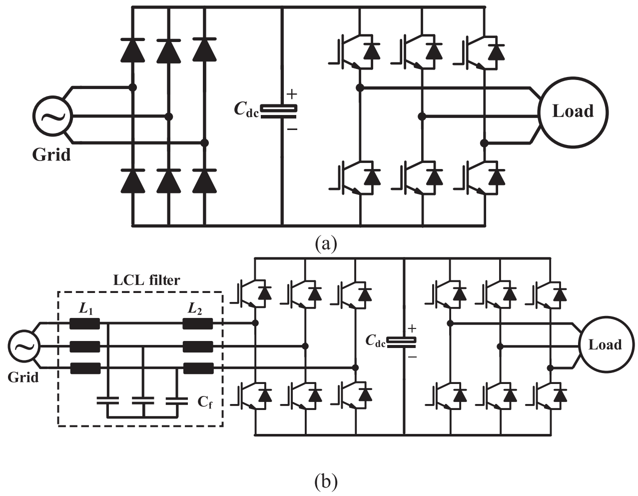


Fig. 2. A power converter with: (a) a diode rectifier, and (b) an Active Frond End topology.

generate. The resonant frequency and behaviour of the converter can be affected by grid impedance and characteristics. Therefore, the grid behaviour (impedance and resonant frequencies) needs to be analyzed with different penetrations of power electronic converters in order to find a proper penetration limit of such converters for residential, commercial, and industrial regions. The second stage is to estimate and identify harmonics emissions (measuring voltage) based on the penetration limit of the converters in grids. The existing commercial converters based on AFE topology generate harmonics mainly above 2 kHz, which are very diverse in frequency and magnitudes due to their modulation strategy, control and LCL filter size. Therefore, the variable phase-angle and magnitude of harmonics within this frequency range needs to be investigated based on statistical approaches. IEC 61000-3-6 defines summation laws to simplify the net harmonic emission generated by a number of systems such as power electronics equipment [24].

Overall, the emerging voltage and current harmonics in the range 2–9 kHz from different sources like motor drives, filters or power generators like PV and wind could impact different aspects of the distribution system operation. One of the most critical assets of the power system, which is the transformers, could also be profoundly affected in a distorted grid. Consequently, it is vital to evaluate the impact of high-frequency harmonics on the transformer using available indices, which is the main subject of this paper.

3. Mathematical models of transformer losses affected by low-order (0–2 kHz) and high-order (2–9 kHz) harmonics

Transformer losses consist of no-load and load losses. No-load (or core) losses mainly depend on voltage harmonics [14,25–27]. On the other hand, load losses are mainly related to current harmonics (fundamental and other orders). Based on IEEE std. C57.110 [12], load losses (P_{LL}) under the presence of current harmonics can be calculated from (1)-(4).

$$P_{LL} = P + P_{EC} + P_{OSL} \tag{1}$$

$$P = I^2R \tag{2}$$

$$P_{EC} = P_{EC-R} * \sum_{h=1}^{max} \left(\frac{I_h}{I_R}\right)^2 h^2 \tag{3}$$

$$P_{OSL} = P_{OSL-R} * \sum_{h=1}^{max} \left(\frac{I_h}{I_R}\right)^2 h^{0.8} \tag{4}$$

Where P is the ohmic loss, P_{EC} is the winding eddy current losses, P_{EC-R} is the rated winding eddy current losses, I_h is the rms current at harmonic h, I_R is the rated rms current, P_{OSL} is the other stray losses, and P_{OSL-R} is the rated other stray losses. Parameters F_{HL} and K_f are then defined as given in (5) and (6).

$$F_{HL} = \frac{P_{EC}}{P_{EC-O}} = \frac{\sum_{h=1}^{max} {I_h}^2 h^2}{\sum_{h=1}^{max} {I_h}^2} \tag{5}$$

$$K_f = \sum_{h=1}^{max} {I_h}^2 h^2 = \sum_{h=1}^{max} \left(\frac{I_h}{I_R}\right)^2 h^2 \tag{6}$$

Where $P_{EC\text{-}O}$ is the winding eddy-current loss at the measured current and the power frequency. In transformer engineering practice, it is notable that large safety margins are incorporated due to the limited accuracy of calculations [28]. Authors in [28] have proposed to use thermal-hydraulic network modelling to establish design tools, whereas the methods used in standards deem to be conservative. From (5) and (6), the relationship between K_f and F_{HL} can be driven as given in (7).

$$K_f = \left(\frac{\sum_{h=1}^{max} {I_h}^2}{I_R^2}\right) F_{HL} \tag{7}$$

In (7), the maximum frequency is considered the highest significant harmonic number based on IEEE C57.110 [12], which is normally considered 2 kHz in practical applications. Hence, the maximum output current (I_{max}) in pu can be calculated based on [6] as given in (8), based on EN/CENELEC [29] as (9), and for dry-type transformer [12] as (10).

$$I_{max} = \sqrt{\frac{R_{DC} + R_{EC-R} - (\Delta P_{Fe} + \Delta P_{OSL})/{I_R}^2}{R_{DC} + F_{HL} R_{EC-R}}} \tag{8}$$

$$I_{max} = \left(\sqrt{1 + \frac{P_{EC-R}}{1 + P_{EC-R}} \left(\frac{I_1}{I}\right)^2 \sum_{h=2}^{max} h^q \left(\frac{I_h}{I_1}\right)^2}\right)^{-1} \tag{9}$$

$$I_{max} = \sqrt{\frac{1 + P_{EC-R}}{1 + F_{HL} P_{EC-R}}} \tag{10}$$

Where R_{DC} is the DC resistance of winding, R_{EC-R} is the additional resistance caused by eddy current at rated frequency, ΔP_{Fe} is the effect of harmonics on iron-core loss, and ΔP_{OSL} is the effect of harmonics on other stray losses. Typical values for q in (9) are 1.7 for transformer with round or rectangular cross-section conductors in both windings or 1.5 for the ones with foil-wrapped windings. Maximal operation capacity (S_M) and reduction in apparent power rating (RAPR) are as given in (11) and (12) respectively, where S_R indicates transformer rated apparent power [6].

$$S_M = I_{max}(pu) * S_R \tag{11}$$

$$RAPR = \frac{S_R - S_M}{S_R} * 100\% = [1 - I_{max}] * 100\% \tag{12}$$

RAPR in (12) is used to identify the derating requirement of the transformer. In the next section, (6) is used to calculate K_f values from simulation results and practical measurements.

4. Impact of harmonics on transformer Kf value

It can be seen from (6) that K_f depends on fundamental current and low- and high-frequency current harmonics. That equation can be rewritten as given in (13) to separate these three contributing factors.

$$K_f = \left(\frac{I_1}{I_R}\right)^2 + Kf_{Lf} + Kf_{Hf} \tag{13}$$

Where Kf_{Lf} and Kf_{Hf} represent the contributions of low-frequency (100–2000 Hz) and high-frequency (2–9 kHz) current harmonics on K_f and can be calculated from (14) and (15) respectively.

$$Kf_{Lf} = \left(\frac{1}{I_R}\right)^2 \sum_{h=2}^{40} h^2 I_h^2 \tag{14}$$

$$Kf_{Hf} = \left(\frac{1}{I_R}\right)^2 \sum_{h=41}^{180} h^2 I_h^2 \tag{15}$$

As high-frequency harmonics could highly impact the level of K_f, defining and calculating Kf_{Lf} and Kf_{Hf} can help to compare the contribution of low- and high-frequency harmonics on K_f. On the other hand, the summation of Kf_{Lf} and Kf_{Hf} represents the overall contribution from current harmonics besides the fundamental component. Consequently, another parameter called Kf_{Tot} to evaluate this overall contribution can be defined as given in (14).

$$Kf_{Tot} = Kf_{Lf} + Kf_{Hf} \tag{16}$$

In this section, these three parameters are evaluated in a modelled distribution system using simulation results, and in three practical systems using field measurement data.

4.1. K_f values extracted from a simulated distribution system

A distribution system has been modelled in MATLAB Simulink, as shown in Fig. 3. The system consists of an MV/LV transformer and five subsystems each with 20 different power electronic converters and one linear load (series RL). Thus, there are a total of 100 power electronic converters and five linear load units in the system. Electrical characteristics of the 11 kV/ 415 V transformer, line impedance, and total system linear load (series RL) are presented in Table 1. The linear load units are distributed in all five subsystems with similar series R (1.56 Ω) and L (2.4 mH) units connected between line and neutral in three phases.

Similarly, characteristics of AC-DC converters in each subsystem are presented in Table 2. These converters consume a total power of around 230 kW.

For the simulated system, Kf_{Lf}, Kf_{Hf}, and Kf_{Tot} values were calculated for different loading levels using (14)-(16) as given in Table 3.

As the power electronic converters utilized diode rectifiers, high-frequency harmonics generated by these power converters are very low in this system, which leads to negligible Kf_{Hf} in all loading conditions. However, the outcomes could have been different if the power electronic converter had fast switching devices operating at a high-frequency switching.

The time-domain current waveform of the transformer and variation of Kf_{Lf} with power electronic converters loading levels are plotted in Fig. 4 (a) and (b). This figure indicates an almost linear increase in Kf_{Lf} when the converter loading level goes higher. It can be seen that at full converter loading, Kf_{Lf} has a value of 0.85, which increases overall K_f to 1.85 from (13). Knowing that $K_f = 1$ in the case of only fundamental current is present, the impact of harmonics on K_f in this simulation study is 85% increase, which is considerable. Consequently, it is important to measure and take into account the impact of current harmonics when evaluating the transformer loading condition. In the next section, Kf values calculated from field measurements are presented and analyzed.

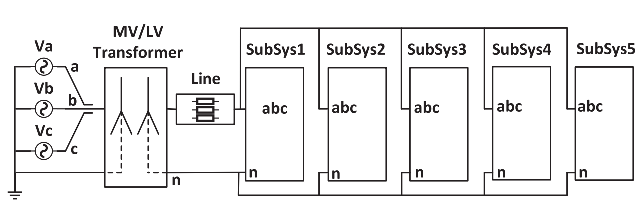


Fig. 3. Substation with MV/LV transformer, line section, and five subsystems modelled in MATLAB Simulink.

Table 1
Transformer, line, and linear load characteristics of the simulated distribution system.

Parameter		Value
Transformer	S_{rated} (kVA)	700
	$I_{rated,LV}$ (A)	972
	leakage reactance (pu)	0.04
	leakage resistance (pu)	0.02
Line	Inductance (pu)	0.01
	Resistance (pu)	0.01
Linear load	total power (kVA)	500
	power factor	0.9

Table 2
Characteristics of AC-DC converters in each subsystem.

Subsystem	Choke type	Power (kW)	Number of units
1	DC	1.0	20
2	DC	2.2	20
3	DC	2.2	20
4	AC	2.2	20
5	AC	4.0	20

Table 3
K_f values for different loading levels of the AC-DC converters.

Loading level (%)	Kf_{Lf}	Kf_{Hf}	Kf_{Tot}
25	0.31	0.1×10^{-4}	0.31
50	0.48	0.8×10^{-4}	0.48
75	0.65	1.6×10^{-4}	0.65
100	0.85	2.0×10^{-4}	0.85

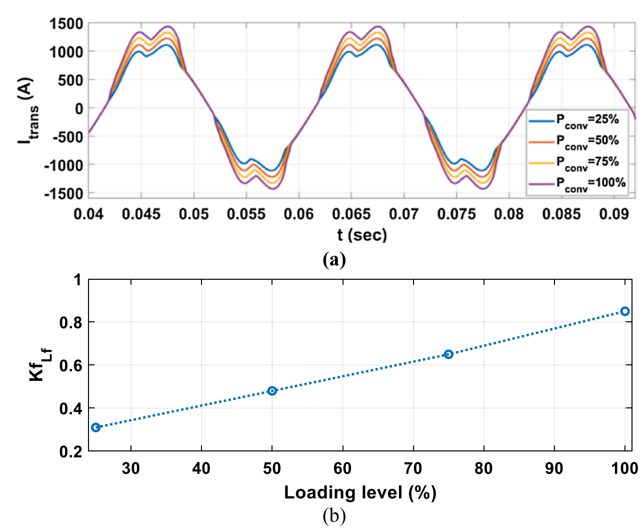


Fig. 4. Different transformer loading levels by changing AC-DC converters loads: (a) time-domain waveform, and (b) variation of Kf_{Lf}.

4.2. K_f values based on field and lab measurements

To evaluate Kf_{Lf}, Kf_{Hf}, and Kf_{Tot} values in practical conditions, current harmonics in both low- and high-order frequency ranges have been measured at three different locations. The relevant plots calculated based on (14)-(16) for these three locations are presented in this section. The first location (system 1) is a commercial area, where a PV installation of 338 kW is connected to an MV/LV transformer rated 750 kVA as shown in Fig. 5 (a) and (b). Other loads in nearby buildings are also connected to the LV side of the transformer. Two power quality meters are connected to the LV side of this transformer, where they

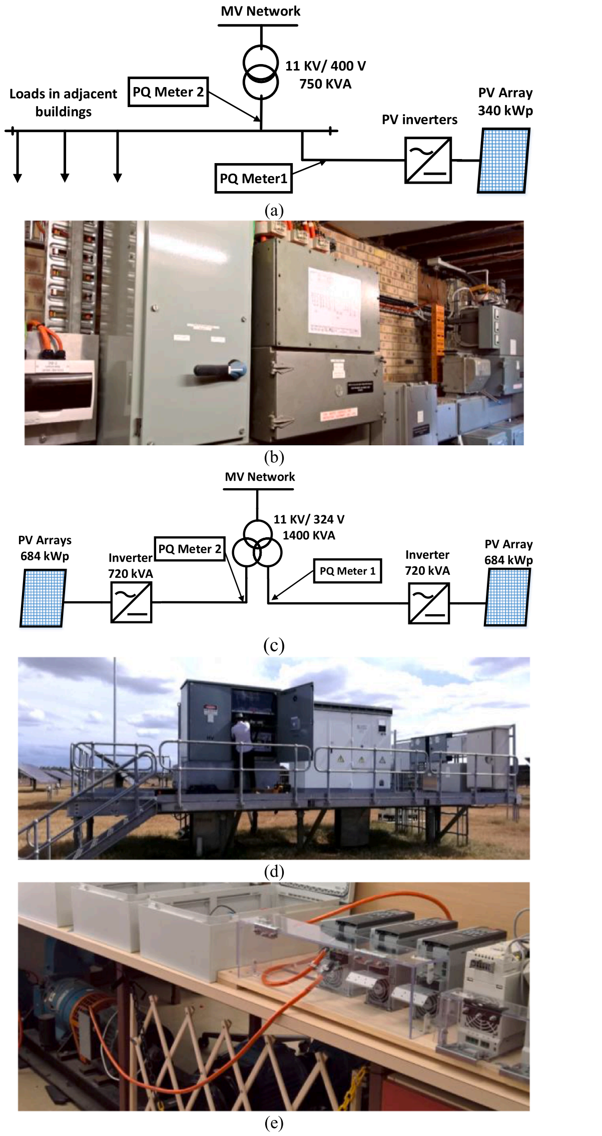


Fig. 5. Measurement locations at system 1: (a) line diagram of the network, (b) field location inside the substation room, and system 2: (c) line diagram of the network and (d) field location (transformer and PV inverters), (e) lab setup for conventional motor drive.

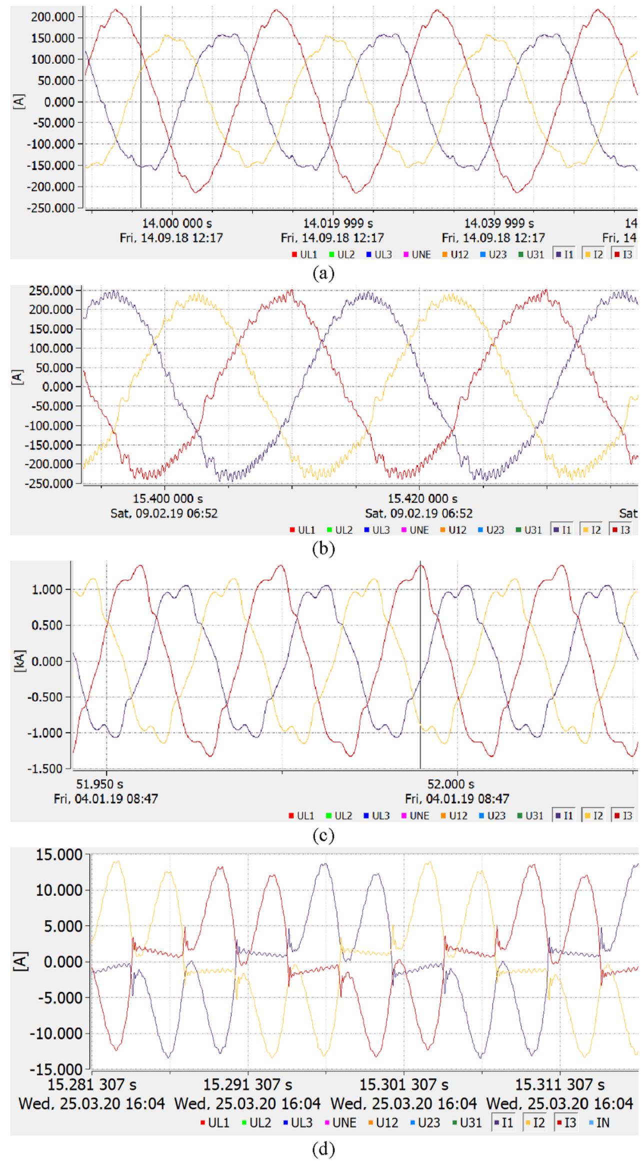


Fig. 6. Time-domain three-phase current waveforms captured for: (a) system 1 (transformer side), (b) system 2, (c) system 3, and (d) system 4.

measure PV installation and transformer total currents, as shown in Fig. 5 (a). These devices are equipped with Rogowski coils to measure current harmonics up to 20 kHz with an accuracy of 1.5% of the measured value. The second location (system 2) uses a 1400 kVA transformer, which has the LV side split into two 700 kVA windings, as shown in Fig. 5 (c). Two large-size PV inverters (720 kVA) are connected to the LV windings. In contrast to system 1, all the energy generated by the PV is exported to the MV network through the transformer and there is no other load between the PV and transformer. The third location (system 3) is an industry site with no PV units and is given here for comparison with the systems with PV. A further measurement has been performed in the lab to evaluate the current harmonic contents of a motor drive and their impact on K_f. This lab setup, which is called system 4 in this paper, is shown in Fig. 5 (e). The DC-choke drive is connected to a 5.5 kW induction motor with a DC-motor coupled to its shaft as a load. Rear-end inverter of the drive operates at 3 kHz switching frequency.

Time-domain three-phase current waveforms for the four systems are presented in Fig. 6.

Kf_{Lf}, Kf_{Hf}, and Kf_{Tot} values calculated from (14)-(16) are shown in Fig. 7(a)-(d) for all four systems during measurement periods (several days for systems 1–3). Values of I_R are calculated 1041.7A for the transformer side of system 1, 1127 A for system 2, 900 A for system 3, and for the lab setup (system 4) it is 7.0 A. It should be noted that as the available data for 2–9 kHz are in 200 Hz steps, each data point is the RMS of four values with 50 Hz steps.

The results for system 4 from Fig. 7 (d) show very large values for all K_f components (Kf_{Lf} = 14.4, Kf_{Hf} = 13.3, and Kf_{Tot} = 27.7) due to the high level of harmonics in the motor drive system. Moreover, the amount of contribution by high-frequency components (Kf_{Hf}) is almost as large as the low-order harmonics contribution (Kf_{Lf}) in Kf_{Tot}. These results indicate that in a system with a high level of conventional motor drive loads, K_f could have large values, which adversely affects the operation of the local transformer. On the other hand, as can be seen from Fig. 7 (a) and (b), there are regular patterns for all K_f values in the measurement periods. However, some transients can also be detected. During these transient times, the parameters are sometimes much more substantial than normal operating conditions. In system 2, transients for Kf_{Hf} up to 40 can be observed during the 3rd of Feb 2019. The conditions causing these transients are not always clear and well-understood.

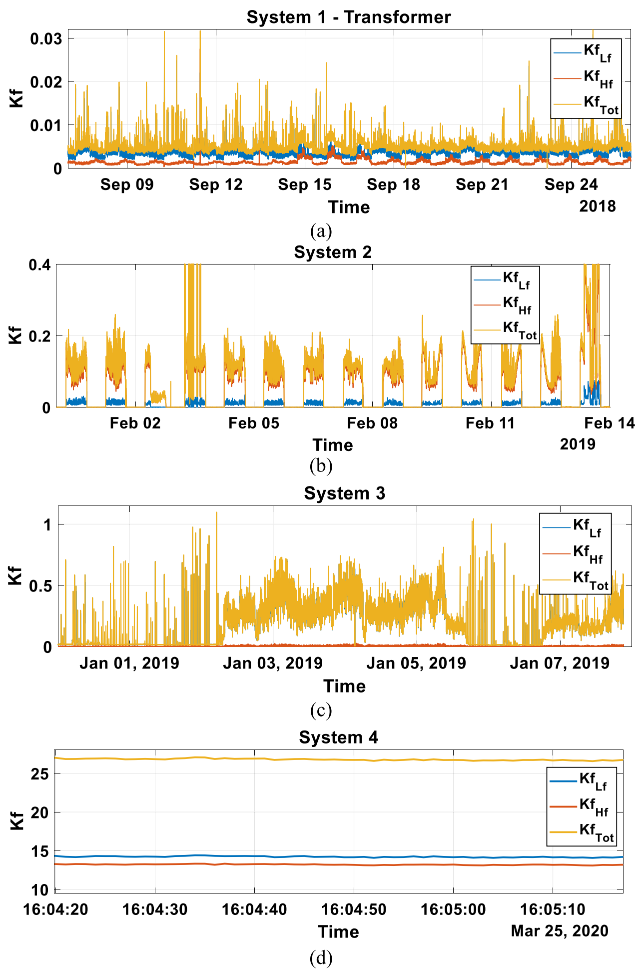


Fig. 7. Kf_{Lf}, Kf_{Hf}, and Kf_{Tot} values for the whole measurement period: (a) system 1 transformer side, (b) system 2, (c) system 3, and (d) system 4.

Moreover, if they become worse and more frequent over time due to changes in network conditions, like connecting more inverters to the system and ageing of protection and other equipment, they could have a more significant effect on the transformer. Consequently, the results are analysed here in two categories of steady-state conditions and transient times as follows.

4.3. K_f plots for steady-state conditions

To compare the impact of low- and high-frequency current harmonics on K_f, the ratio of Kf_{Hf}/Kf_{Lf} and the level of individual harmonic component contribution (calculated from 95 percentile values for systems 1–3 and average values for system 4) are shown in Fig. 8. The measurement duration is a sample 24-hour period for system 1–3 and 1 min for system 4. As can be seen from Fig. 8 (a) in the transformer side of system 1, where the current is the addition of PV and connected load currents, Kf_{Lf} is mainly higher than Kf_{Hf}. In this system, there is a possibility that current harmonics generated by PV and load units cancel out each other. This could be especially the case in the higher frequency range, where the harmonics phase-angles have a high variation. Fig. 8 (b) shows that, in the low-frequency range, mainly odd current harmonics contribute to K_f level due to their higher magnitude, with the most significant contributions are from 3rd, 7th, and 11th harmonics. This figure also indicates that the contribution of high-frequency current harmonics to the K_f value is mainly up to 5.5 kHz, while higher frequency components are negligible.

Fig. 8 (c) for system 2 shows that Kf_{Hf} is always significantly higher

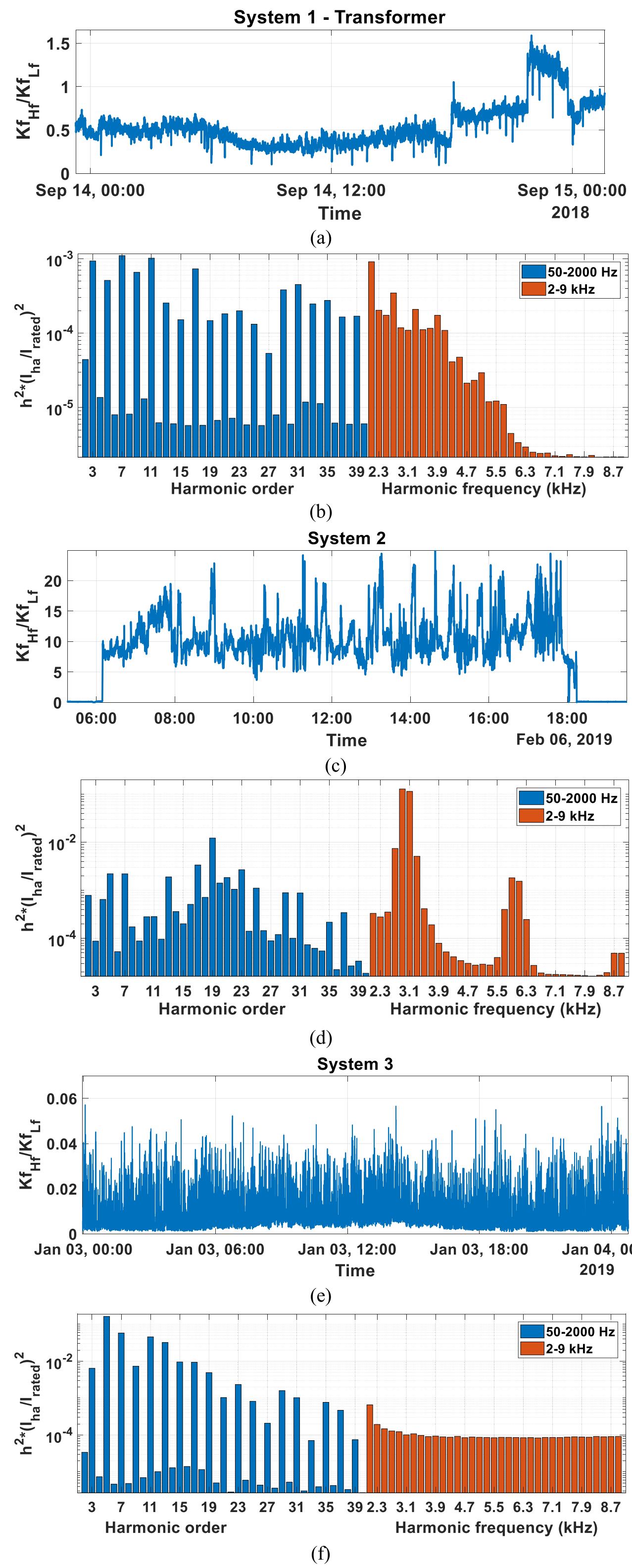
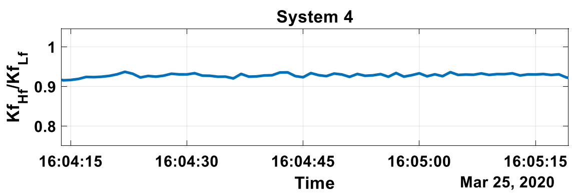
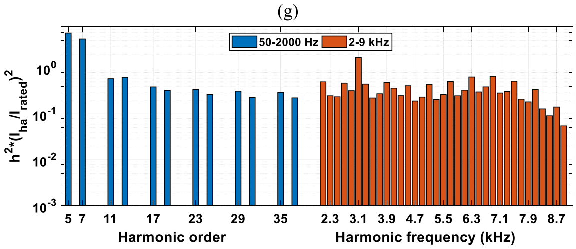


Fig. 8. Kf measurements: (a), (c), (e), and (g): the ratio of high-frequency over low-frequency terms for systems 1–3, (b), (d), (f), and (h): distribution of low- and high-frequency components for systems 1–4.

than Kf_{Lf} (5–25 times) during PV power generation. Consequently, if only low-frequency harmonics were considered to calculate K_f and high-frequency harmonics were ignored, there would be a significant underestimation of the exact level of K_f. This underestimation could result in miscalculation of current harmonics impact on the transformer



(g)



(h)

Fig. 8. (*continued*)

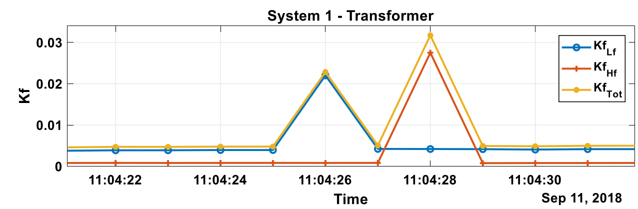
temperature and lifetime. The difference between the impact of low- and high-frequency components can be better understood by looking at Fig. 8 (d), where there are large components in 2.9 kHz and 3.1 kHz frequencies, which are around ten times larger than the highest low-frequency component (19th order). There are also high contributions toward K_f around 6 kHz frequency. The difference between system 1 and 2 is that the PV generation in system 1 comes from several small-size inverters connected in parallel, whereas in system 2, there is one large-size inverter for all the PV generation. The measurements show that the switching frequency of small-size inverters in system 1 is in the range 16–17 kHz, whereas in system 2, it is around 3 kHz, which could be a source of difference in Kf_{Hf} levels measured in these systems. It can be seen from Fig. 8 (e)-(f), Kf_{Hf} is very small in system 3, which indicates that current harmonics in the range 2–9 kHz are negligible. In the next section, K_f values during observed transient incidents in systems 1 and 2 are presented. Finally, Fig. 8 (g) indicates that Kf_{Hf} is around 92% of Kf_{Lf} for the measurement performed in system 4. As it could be expected from a conventional motor drive, the main contribution in low-order harmonics is from odd harmonics except for triplen ones, with highest from 5th and 7th due to their larger magnitudes. At higher frequency in this system, the main contribution is from around 3 kHz, which is the switching frequency of the rear-end inverter.

4.4. Kf plots in transient times

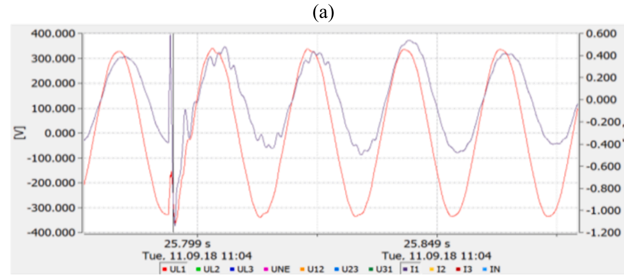
It can be seen from Fig. 7 (a) for system 1 transformer side measurements that there are some transient incidents where Kf_{Tot} reached around 0.03, which is higher than other times (0.01–0.02). K_f values around the transient time are plotted in Fig. 9 (a). The measurement device has also recorded scope image data at the time of the incident, which is shown in Fig. 9 (b). This figure shows a voltage sag incident around the same time, which was quickly recovered. However, the current waveform has some fluctuations for several cycles, which affected the harmonic levels at that time.

In system 2, transients can be seen on Feb 3, as shown in Fig. 7 (b). Kf_{Lf}, Kf_{Hf}, and Kf_{Tot} values around one of these transients are shown in Fig. 10. There is no scope recording data available for this period. However, comparing the values of Kf_{Hf} and Kf_{Lf} indicates that a significant level of high-frequency current harmonics created Kf_{Hf} values up to 40 during transient times in this system. As can be seen from Fig. 7 (c), no large transient happened during the measurement period for system 3.

It needs to be mentioned that as the studied transients in this section are in the timeframe of several seconds and momentary, they cannot considerably increase the transformer temperature due to its thermal inertia. However, such current surges impinge a mechanical force on



(a)



(b)

Fig. 9. Transient time in system 1, transformer side: (a) Kf values, and (b) recorded voltage and current waveforms in phase a.

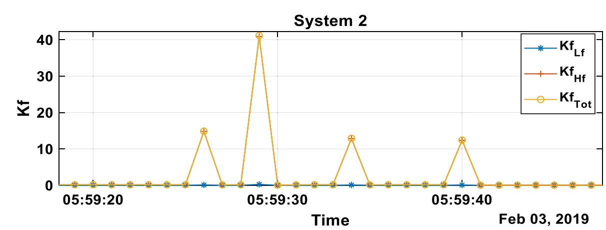


Fig. 10. Kf values around one of the transient times in system 2.

the winding structure of a transformer. If the winding is distorted, sections of the insulation can become overstressed, degrade, and eventually fail. An analysis of transformer failure modes found that approximately half of the fleet fail is in this way [30]. It has also been mentioned in [30] that about half of transformer failures are due to the mechanical shock of a through-fault. As utilities frequently measure only rms current, it is recommended that the number of through-faults and their intensity be also counted to have a more realistic and accurate estimation of transformer lifetime. In this way, it can be identified if a transformer is at risk of premature failure due to high-intensity current transients, and, consequently, remedial actions could be considered.

In the next section, the impact of current harmonics on the temperature rise and lifetime of the transformer in system 2 is investigated as a case study.

5. Impacts of current harmonics on transformer losses and accelerated ageing factor

The standard IEEE C5791-2011 [31] uses an accelerated ageing factor (F_{AA}), which equals 1 at 110 °C, to quantify the effect of temperature on insulation. However, whereas this loading guide is for transformers containing thermally upgraded paper, the target transformer use standard Kraft paper, and so F_{AA} was baselined at 98 °C following IEC 60076-7:2018 [32]. These transformers were filled with natural ester, and so ageing coefficients reflective of this fluid were used.

In the case of system 2, the transformer was operated at 1.4p.u. load, and thus the effect of current harmonics at this high load was investigated. The IEEE std C57.110-2018 [12] provides a method to calculate a harmonic multiplier for winding eddy currents and stray losses. In this method, assumptions are used to estimate the effects of low frequencies on the harmonic multiplier. A discussion is given on correcting the harmonics loss factor for high frequencies such as calculating the skin effect from the strand dimension [12]. However, strand dimension data might not be readily available for a transformer. A corrected loss is determined and is then used to calculate the effect on temperature. The rate of ageing of paper doubles approximately every 6 °C rise in temperature based on IEC 60076-7:2018 [32]. The increase in winding eddy losses is related to the square of the harmonic current multiplied by the square of the frequency. Consequently, plotting $(I_h/I_R)^2 \times h^2$ for each harmonic will show its respective influence on the winding current, compared to the fundamental current. This has already shown in the previous section, Fig. 7 (d), where the effect of each harmonic, benchmarked against the rated current. Notably, there are two peaks at the 58th and 62nd harmonics, which have a combined eddy current effect of 33% of the fundamental current. Since these harmonics are higher than the 50th harmonic, they may be unnoticed by a transformer user, yet their magnitude is significant. The approaches to ascertain the impact of temperature on the insulation life are:

- Calculate the losses due to current harmonics, compared to the normal load loss that would result from the load if no harmonics were present.
- Calculate the expected increase in temperature caused by the harmonics.
- Calculate how the residual life of the insulation paper has been affected.

The transformer in system 2 has the characteristics: 1.4/0.7/0.7 MVA arranged as two secondary windings at 11/0.324/0.324 kV. $R_1 = 0.827\ \Omega$ and $R_2 = 0.0011\ \Omega$, and $I_{1\text{-}R} = 73$ A and $I_{2\text{-}R} = 2{,}254$ A ($I_{2\text{-}R}$ corresponds to both the two LV windings being operated at rated load, where each winding has a rated current of 1,127 A). Total stray loss can then be calculated from (17), which equals 3945 W.

$$P_{TSL-R} = P_{LL} - 1.5 \times (I_{1-R}^2 \times R_1 + I_{2-R}^2 \times R_2) \tag{17}$$

Following IEEE C5791-2011 [31], the winding eddy loss can be calculated as given in (18).

$$P_{EC-R} = P_{TSL-R} \times 0.2 = 789W \tag{18}$$

Where the relevance of 0.2 is that it is an estimate of the LV winding eddy loss, as a ratio to the total stray loss. Other stray losses can be calculated as given in (19).

$$P_{OSL-R} = P_{TSL-R} - P_{EC-R} = 3156W \tag{19}$$

The impact of harmonics on the eddy currents of the transformer winding is calculated using the harmonic loss factor F_{HL} determined by (5). The harmonic distribution was processed, with the determination of the various factors required for the heating calculation given in Table 4. $F_{HL-Eddy}$ is determined by dividing column 2 in Table 4 by column 1, and F_{HL-STR} is calculated by dividing column 3 by column 1, which are 1.391 and 1.004, respectively. Table 5 shows the total load loss with no harmonic and corrected with the observed harmonics. It can be seen from this table that when harmonics are present, there is an increase in losses of 2.1%.

The ultimate top-liquid temperature rise for this loading condition is calculated as given in (20).

$$\theta_{TO} = \theta_{TO-R} \times \left(\frac{P_{LL} + P_{NL}}{P_{LL-R} + P_{NL}}\right)^{0.8} \tag{20}$$

Table 4
Calculation of harmonic loss factors for winding eddy (column 2) and stray (column 3) losses.

Column	1	2	3
h	$\left(\frac{I_h}{I_1}\right)^2$	$\left(\frac{I_h}{I_1}\right)^2 h^2$	$\left(\frac{I_h}{I_1}\right)^2 h^{0.8}$
Σ	1.0005	1.392	1.004

Table 5
Calculation of Losses.

Loss type	Rated loss (W)	Load loss (W)	Harmonic multiplier	Corrected loss (W)
No-load	1286	1286		1286
Load loss	2[†] × 14,216			
Contribution towards load loss				
I^2R	2 × 10,271	2 × 13,903		2 × 13,903
Winding eddy	2 × 789	2 × 1068	1.391	2 × 1486
Other stray	2 × 3156	2 × 4272	1.004	2 × 4289
Total	29,717	39,770		40,639

[†] Denotes load loss for each of the two LV windings.

where θ_{TO-R} is given in the product datasheet as 60 °C. The ultimate winding hotspot rise is given by (21).

$$\theta_g = \theta_{g-R} \times \left(\frac{I_2^2R + P_{EC-Corrected} \times 0.2 \times 4}{P_{I^2R} + P_{EC}}\right)^{0.8} \tag{21}$$

Where 0.2 reflects that the LV winding eddy current losses are 20% of the total losses, and 4 is from the assumption that the maximum eddy loss at the hottest spot region is four times the average eddy loss.θ_{g-R} is assumed to be 18 °C in accordance with the IEC 60076-7:2018 [32]. I_2^2R is then calculated from (22) and (23).

$$I_2^2R = I_{2-R}^2 \times P_{LL}(pu)^2 \tag{22}$$

$$I_{2-R}^2R = 1.5 \times I_{2-R}^2 \times R_2 \tag{23}$$

where 1.5 denotes that the transformer is three-phase. Under the observed conditions θ_g is 77.1 °C with harmonics and 75.8 °C without, which leads to the hottest spot conductor rise over ambient of 100.2 °C with harmonics versus 98.4 °C without. Since the relative rate of paper ageing doubles for every increase of 6 °C, an increase of 1.7 °C will accelerate this rate by 22%. Consequently, it is recommended here that the transformer owner must consider this accelerated ageing in life remaining calculations when sizing a transformer.

6. Discussion

The main subject of this work was to present the possibility of a significant impact of higher frequency current harmonics on the transformer loss. Presented practical results for system 2 indicates that the impact of current harmonics at frequencies higher than 2 kHz could be even more severe on transformer temperature than lower frequency harmonics. This has been confirmed using K_f method.

The possibility of a significant impact of higher frequency current harmonics on the transformer is higher at systems with a stable source of high-frequency current harmonics. Although such sources do not exist at systems 1 and 2, system 3 has a large-size PV inverter with switching frequency around 3 kHz. Consequently, due to the existence of sources of current harmonics at higher frequencies in distribution systems, authors recommend that their impact on transformers needs to be evaluated. It has also been shown in this paper that higher frequency harmonics could reduce the transformer insulation lifetime by around 22% in the case of system 2. This example further confirms the importance of considering high-frequency harmonics when evaluating the transformer temperature rise and insulation aging.

Based on practical measurements, it has also been shown that there have been transient large high-frequency current harmonics in a system with large-size PV inverters. Although these transients cannot change the overall temperature of the transformer due to their short-term nature, they can impose a mechanical force on the winding structure of a transformer. If the winding is distorted, sections of the insulation can become overstressed, degrade, and eventually fail in an extended time.

7. Conclusions

In this work, the impact of current harmonics on K-factor (K_f) value of the distribution transformer is investigated. Several simulation results and practical measurements are used to evaluate this factor in different systems and loading conditions. To have a more in-depth analysis of this factor and comparison between the impact of fundamental current, low-frequency (100–2000 Hz), and high-frequency (2–9 kHz) current harmonics, K_f is considered as the summation of three factors: fundamental factor, Kf_{Lf} (contribution of low-frequency harmonics), and Kf_{Hf} (contribution of high-frequency harmonics). Moreover, the addition of Kf_{Lf} and Kf_{Hf} has been defined as Kf_{Tot}, which indicates the overall contribution of current harmonics in K_f. It has been found that for one case when a large-size PV inverter is connected to the transformer, the contribution of current harmonics higher than 2 kHz on K_f value (Kf_{Hf}) is significant in a way that sometimes it is 15–25 times higher than low-order harmonics (Kf_{Lf}). The importance of this finding is that normally calculation of K_f is performed by considering harmonics only up to 2 kHz.

Using field measurements for a practical system, it is also shown that high-frequency harmonics have resulted in a 33% increase in winding eddy losses, which would not be apparent if a utility only records harmonics up to the 50th. This higher power loss due to higher frequency harmonics causes 22% acceleration in ageing during the time of interest. Consequently, it is recommended to include sections in IEEE or IEC standards on HF harmonics with suggestions about a more detailed approach to modelling due to skin effect, and using Finite Element Modelling, FEM. Currently, IEC TC77A, WG1 and WG8 are preparing compatibility levels for harmonics within 2–9 kHz (WG1) and 9–150 kHz (WG8) based on communication and signalling errors in the power network. However, the impact of high-frequency harmonics on transformers is one of the main concerns which should be considered in these standards. This may make modelling a utility transformer more challenging as adequate design information might not be available. However, more research required to extend the IEEE and IEC standards also to include high-frequency harmonics.

CRediT authorship contribution statement

Jalil Yaghoobi: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data curation, Writing - original draft, Visualization. **Abdulrahman Alduraibi:** Methodology, Software, Validation, Formal analysis, Data curation, Writing - original draft, Visualization. **Daniel Martin:** Conceptualization, Methodology, Formal analysis, Investigation, Resources, Data curation, Writing - original draft. **Firuz Zare:** Conceptualization, Methodology, Formal analysis, Investigation, Resources, Writing - review & editing, Visualization, Supervision, Project administration, Funding acquisition. **Daniel Eghbal:** Conceptualization, Writing - review & editing, Visualization, Supervision, Project administration, Funding acquisition. **Rizah Memisevic:** Conceptualization, Writing - review & editing, Visualization, Supervision, Project administration, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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